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[]: Question no 1:

```
[8]: person = {"name": "sana", "age": 19, "city": "NYC", "hobbies": ["coding",  
↪ "hiking"]}  
print(person)
```

```
{'name': 'sana', 'age': 19, 'city': 'NYC', 'hobbies': ['coding', 'hiking']}
```

[]: Question No 2:

```
[13]: text = "apple banana apple cheery banana apple"  
words = text.split()  
freq = {word: words.count(word) for word in set(words)}  
print(freq)
```

```
{'banana': 2, 'cheery': 1, 'apple': 3}
```

[]: Question No 3:

```
[11]: inv = {"apple": 50, "banana": 20}  
inv["oranges"] = 30  
inv["apple"] -= 5  
print(inv)
```

```
{'apple': 45, 'banana': 20, 'oranges': 30}
```

[]: Question No 4

```
[19]: orders = ["electronic", "book", "electronic", "book", "clothing", "electronic"]  
most_popular = max(set(orders), key=orders.count)  
print("Most_Popular:", most_popular)
```

```
Most_Popular: electronic
```

[]: Question No 5:

```
[29]: grades = {"math": 88, "science": 75, "histort": 73}
avg = sum(grades.values()) / len(grades)
print(f"average grad: {avg: .2f}")
```

average grad: 78.67

[]: Question No 6

```
[38]: stock = {"pen":{"qty": 10, "price": 1.5, "notebook": 3, "pen": 5.0}}
stock["pen"]["qty"] -= 7
total_val = sum(p["qty"] * p["price"] for p in stock.values())
low_stock = [n for n, p in stock.items() if p["qty"] < 5]
print(f"total.value: {total_val}")
print(f"low stock items: {low_stock}")
```

total.value: 4.5

low stock items: ['pen']

[]: Question No 7:

```
[47]: book = {"alex": [80, 90, 70], "sum": [95, 85, 100]}
avgs = {name: sum(g)/len(g) for name, g in book.items()}
sorted_st = sorted(avgs.items(), key=lambda x: x[1], reverse=True)
print("Rankings:", sorted_st)
```

Rankings: [('sum', 93.33333333333333), ('alex', 80.0)]

[]: Question No 8:

```
[52]: catalog = {"laptop": 1000, "mouse": 50}
cart = {"laptop": 1, "mouse": 2}
subtotal = sum(catalog[item] * qty for item, qty in cart.items())
total = subtotal * 1.10
print("subtotal: {subtotal}")
print(f"Total with tax: {total}")
```

subtotal: {subtotal}

Total with tax: 1210.0